

Si Yudong

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☐ Education

Tongji University

2019.09 - 2022.03

Nantong University

2014.09 - 2018.03

☐ Work Experience

AI Framework Engineer (Triton Compiler)@Intel.

2022.07 - 2025.07

Triton Compiler XPU Backend Development

• Feature Implementation

- Implemented global scratch memory addressing in backend launcher
- Built SPIR-V → Level Zero → SYCL AOT compilation pipeline, facilitating low-level issue reproduction.
- Created Intel specific Pass tritonintelgpu-rewrite-stack-ptr to optimize shared local memory (SLM) pointer handling.
- Integrated __spirv_RoundFToTF32INTEL SPIRV extension for enhanced FP32 precision in tl.dot DPAS lowering.

• Performance Optimization

- Established Triton-XPU's first benchmark system covering Softmax/GEMM/FA key kernels, with extensible support for vendor libraries (XeTLA/CUTLASS/oneDNN).
- Landing optimization passes like: Coalesce, AcceletateMatmul, RemoveLayoutConversion, Pipeline / Prefetch, Swizzling etc.
- Optimized critical kernels (e.g. GEMM) and achieved >90% performance of Intel's XeTLA library

• Bug Fixes

- Resolved 35+ High-priority compiler backend issues including: performance regressions, Layout propagation failures in IR passes, ut issues and so on.

PyTorch Ecosystem Optimization

• CI/CD Innovation

- Designed AWS Xeon-based Jenkins pipeline for PyTorch Inductor CPU Performance, automated collection of 150+ model performance metrics

☐ Project Experience

Embedded Computer Vision System

• Driver Fatigue Detection System (Jetson Nano and IMX6UL)

- Face detection: Traditional HOG + SVM algorithm
- Key point localization: A 26-point localization model trained based on cascading regression trees
- Fatigue state detection: Implementation of PERCLOS algorithm and Euler angle calculation to identify features such as eye closure, yawning, and head tilting
- Alerting: Development of an operational interface using the QT, integration with in-vehicle LIN ambient lighting for alert functionality

☐ Skills

• **Compiler:** Triton, LLVM, MLIR, SPIRV

• **Heterogeneous Computing:** oneAPI, OpenCL, CUDA, SYCL, Level Zero, ROCm

• **AI Framework:** Pytorch, IPEX

• **AI Infrastructure:** Jenkins, GitHub Actions, Docker, Vtune Profiler

• **Languages:** C/C++, Python, Bash